

CAMPO GRANDE, Brazil

WMO: 836120

Lat: 20.469S

Lon: 54.673W

Elev: 559

StdP: 94.79

Time Zone: -4.00 (SAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.0	10.0	0.0	4.0	16.5	2.2	4.7	19.7	10.1	23.1	9.3	22.3	4.5	130	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.3	35.0	20.2	34.0	20.6	33.0	20.9	24.6	29.9	24.1	29.6	23.7	29.2	5.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.1	19.1	26.6	22.8	18.7	26.4	22.2	18.0	26.0	77.5	29.9	75.9	29.6	74.4	29.3	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.5	7.8	5.5	36.5	1.5	1.5	4.4	37.6	3.6	38.5	2.7	39.3	1.7	40.4
			3.6	25.3	2.0	0.5	2.1	25.7	0.9	26.0	-0.2	26.3	-1.7	26.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.0	25.6	25.7	25.3	24.3	21.7	20.8	21.0	22.9	24.6	25.4	25.5	25.8
	DBStd	3.51	1.55	1.39	1.66	2.68	3.46	3.25	4.06	4.39	4.18	3.04	2.18	1.78
	HDD10.0	2	0	0	0	0	0	0	2	0	0	0	0	0
	HDD18.3	87	0	0	0	3	14	18	26	18	7	1	0	0
	CDD10.0	5120	483	439	473	428	362	324	343	399	437	478	464	489
	CDD18.3	2163	224	206	214	180	119	92	109	159	194	221	214	231
	CDH23.3	17751	1634	1472	1591	1382	775	615	877	1536	2046	2109	1854	1859
	CDH26.7	6919	588	526	567	502	224	140	278	704	1008	946	732	704
Wind	WSAvg	4.0	3.6	3.5	3.3	3.5	3.8	4.0	4.3	4.7	4.7	4.3	4.0	3.7
Precipitation	PrecAvg	1498	223	168	145	100	111	45	42	33	86	136	159	212
	PrecMax	1913	508	314	271	192	268	203	184	122	251	289	299	366
	PrecMin	868	74	47	57	24	7	0	0	0	7	7	20	44
	PrecStd	260	103	63	59	51	75	47	44	37	64	65	72	87
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	33.2	33.3	33.1	31.2	30.2	31.9	34.8	36.9	36.8	34.2	34.0
		MCWB	22.6	22.5	22.6	22.2	20.8	19.0	18.2	18.7	19.2	20.9	22.3	21.9
	2%	DB	32.2	32.1	32.1	31.9	30.1	29.2	30.8	33.2	35.2	35.0	33.1	32.9
		MCWB	22.8	22.6	22.8	21.7	20.6	18.8	17.7	18.2	19.2	21.6	22.0	22.4
	5%	DB	31.1	31.1	31.0	30.9	29.1	28.2	29.5	32.1	34.0	33.2	32.0	31.9
		MCWB	22.7	22.6	22.5	21.3	20.2	18.7	17.8	18.2	19.2	21.8	21.9	22.6
	10%	DB	30.0	30.0	30.0	29.9	28.0	27.1	28.2	30.9	32.8	31.9	30.9	30.8
		MCWB	22.4	22.1	22.1	20.9	19.8	18.6	17.4	17.8	19.1	21.8	21.8	22.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	24.7	25.1	24.1	23.6	22.5	20.8	21.5	23.1	24.3	24.6	24.9
		MCDB	30.1	29.9	30.2	29.5	28.9	26.8	26.6	30.2	31.5	31.4	30.0	29.9
	2%	WB	24.2	24.1	24.3	23.4	22.6	21.3	20.0	20.3	22.3	23.6	23.9	24.2
		MCDB	29.4	29.5	29.4	29.0	27.3	25.5	26.3	29.1	30.5	30.8	29.7	29.4
	5%	WB	23.7	23.6	23.7	22.9	21.6	20.4	19.3	19.5	21.5	23.0	23.4	23.7
		MCDB	28.8	28.9	28.7	28.5	25.7	25.0	25.9	28.4	29.3	30.1	29.1	28.9
	10%	WB	23.2	23.1	23.2	22.3	20.9	19.6	18.5	18.7	20.7	22.4	22.9	23.3
		MCDB	28.2	28.1	28.0	27.2	25.1	24.7	25.4	27.6	27.9	29.0	28.5	28.4
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.0	9.3	9.8	9.6	10.2	11.4	12.2	11.9	10.5	10.0	9.3
		MCDBR	10.7	10.4	10.5	10.8	10.7	11.3	12.6	12.9	13.3	12.2	11.2	10.7
		MCWBR	3.9	3.6	3.8	3.7	4.1	4.3	4.6	4.7	4.4	4.1	4.2	4.0
	5% WB	MCDBR	9.2	9.1	9.2	9.1	9.0	9.6	11.0	12.0	11.5	10.4	9.6	9.2
		MCWBR	3.9	3.7	3.8	3.9	4.1	4.5	4.9	5.1	4.7	4.1	4.0	3.9
Clear-Sky Solar Irradiance	taub	0.392	0.398	0.385	0.365	0.349	0.373	0.370	0.433	0.574	0.490	0.409	0.404	
	taud	2.458	2.439	2.470	2.509	2.506	2.437	2.417	2.190	1.825	2.116	2.361	2.405	
	Ebn at Noon	952	935	923	902	878	832	847	820	743	842	930	942	
	Edh at Noon	121	121	113	101	94	97	101	137	212	165	132	127	
All-Sky Solar Radiation	RadAvg	5.86	5.81	5.59	5.05	4.09	3.80	4.09	4.81	5.07	5.56	5.97	6.00	
	RadStd	0.59	0.32	0.40	0.33	0.36	0.28	0.34	0.31	0.46	0.32	0.46	0.41	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page